

DSA theory mcqs

1st variant: (Time complexity)

1. For a given function $g(n)$, we denote by $O(g(n))$ the set of functions -

- A) $O(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq g(n) \leq c \cdot f(n) \text{ for all } n \geq n_0 \}$
- B) $O(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq g(n) \leq c \cdot f(n) \text{ for all } n \leq n_0 \}$
- C) $O(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq f(n) \leq c \cdot g(n) \text{ for all } n \geq n_0 \}$
- D) $O(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq f(n) \leq c \cdot g(n) \text{ for all } n \leq n_0 \}$

Correct Answer: C

2. For a given function $g(n)$, we denote by $\Omega(g(n))$ the set of functions -

- A) $\Omega(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq c + g(n) \leq f(n) \text{ for all } n \geq n_0 \}$
- B) $\Omega(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq c + g(n) \leq f(n) \text{ for all } n \leq n_0 \}$
- C) $\Omega(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq c \cdot g(n) \leq f(n) \text{ for all } n \geq n_0 \}$
- D) $\Omega(g(n)) = \{ f(n): \exists c, n_0 > 0 \text{ s.t. } 0 \leq c \cdot g(n) \leq f(n) \text{ for all } n \leq n_0 \}$

Correct Answer: C

3. For a given function $g(n)$, we denote by $\Theta(g(n))$ the set of functions -

- A) $\Theta(g(n)) = \{ f(n): \exists c_1, c_2, n_0 > 0 \text{ s.t. } 0 \leq c_1 + g(n) \leq f(n) \leq c_2 + g(n) \text{ for all } n \geq n_0 \}$
- B) $\Theta(g(n)) = \{ f(n): \exists c_1, c_2, n_0 > 0 \text{ s.t. } 0 \leq c_1 + g(n) \leq f(n) \leq c_2 + g(n) \text{ for all } n \leq n_0 \}$
- C) $\Theta(g(n)) = \{ f(n): \exists c_1, c_2, n_0 > 0 \text{ s.t. } 0 \leq c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n) \text{ for all } n \geq n_0 \}$
- D) $\Theta(g(n)) = \{ f(n): \exists c_1, c_2, n_0 > 0 \text{ s.t. } 0 \leq c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n) \text{ for all } n \leq n_0 \}$

Correct Answer: C

2nd variant: (Properties of Data Structures)

1. The 2 dimensional array (Matrix) is
- A) static and sequential data structure
 - B) dynamic and sequential data structure
 - C) static and associative data structure
 - D) dynamic and associative data structure

Answer: C

2. The 1 dimensional array (vector) is
- A) static and sequential data structure
 - B) dynamic and sequential data structure
 - C) static and associative data structure
 - D) dynamic and associative data structure

Answer: C

3. The stack is
- A) static and sequential data structure
 - B) dynamic and sequential data structure
 - C) static and associative data structure
 - D) dynamic and associative data structure

Answer: B

4. The queue is
- A) static and sequential data structure
 - B) dynamic and sequential data structure
 - C) static and associative data structure
 - D) dynamic and associative data structure

Answer: B

5. The linked list is
- A) static and sequential data structure
 - B) dynamic and sequential data structure
 - C) static and associative data structure
 - D) dynamic and associative data structure

Answer: B

6. The table is
- A) static and sequential data structure
 - B) dynamic and sequential data structure
 - C) static and associative data structure
 - D) dynamic and associative data structure

Answer: D

7. The tree is
- A) dynamic and hierarchical data structure
 - B) dynamic and sequential data structure
 - C) static and associative data structure
 - D) dynamic and associative data structure

Answer: A

8. The set is

- A) static and sequential data structure
- B) dynamic and sequential data structure
- C) static and associative data structure
- D) dynamic and without structure data structure

Answer: D

9. The multi-set is

- A) static and sequential data structure
- B) dynamic and sequential data structure
- C) static and associative data structure
- D) dynamic and without structure data structure

Answer: D

10. The sparse tree is

- A) static and sequential data structure
- B) dynamic and sequential data structure
- C) static and associative data structure
- D) dynamic and associative data structure

Answer: C

3rd variant: (Tree, Table and Linked Lists)

1. For self ordering table we can use -
A) linked representation and serial search
B) linked representation and binary search
C) continuous representation and serial search
D) continuous representation and serial search

Answer: A

2. We cannot construct original binary tree from -
A) preorder, inorder and postorder traversal
B) preorder and postorder traversal
C) inorder traversal
D) None of them

Answer: A

3. We can construct expression tree from -
A) prefix, infix and postfix expression
B) prefix and postfix expression
C) infix expression
D) None of them

Answer: B

4. The BST property allows printing keys in sorted order by -
A) preorder tree walk
B) inorder tree walk
C) postorder tree walk

Answer: B

5. The average-case time of searching in a hash-table is
A) $O(\log n)$
B) $O(n)$
C) $O(n^2)$
D) $O(1)$

Answer: D

6. The average-case time of searching in a linked list is
A) $O(\log n)$
B) $O(n)$
C) $O(n^2)$
D) $O(1)$

Answer: B

4th variant: (Red-Black Tree)

1. A red-black tree with n internal nodes has height at most -

- A) $\log_2(n)$
- B) $2 \cdot \log_2(n)$
- C) $\log_2(n+1)$
- D) $2 \cdot \log_2(n+1)$

Answer: D

2. For each node of a red-black tree, all simple paths from node to leaves -

- A) contain the same number of black nodes
- B) contain the same number of red nodes
- C) contain different number of black nodes
- D) contain different number of red nodes

Answer: A

3. Find the FALSE property of this red-black description:

- 1. Every node is either red or black.
- 2. The root is black.
- 3. Every leaf (NIL) is black.
- 4. If a node is red, then both its children are black.
- 5. For each node, all simple paths to leaves have the same number of red nodes.

Answer: 5

4. Find the FALSE property of this red-black description:

- 1. Every node is either red or black.
- 2. The root is black.
- 3. Every leaf (NIL) is black.
- 4. If a node is black, then both its children are red.
- 5. For each node, all simple paths to leaves have the same number of black nodes.

Answer: 4

5th variant: (B-Tree)

1. Nodes have lower and upper bounds in terms of minimum degree $t \geq 2$. Choose the correct property:

- A) Every node other than the root must have at least $2*t-1$ keys.
- B) Every node other than the root may contain at most $2*t-1$ keys.
- C) Every node may contain at most $2*t-1$ keys.
- D) Every node must have at least $2*t-1$ keys.

Answer: C

2. The minimum value of degree of a B-tree is

- A) 0
- B) 1
- C) 2
- D) 3

Answer: C

3. Nodes have bounds in terms of $t \geq 2$. Choose the correct property:

- A) Every node other than the root must have at least $t-1$ keys.
- B) Every node may contain at most $t-1$ keys.
- C) Every node other than the root may contain at most $t-1$ keys.
- D) Every node must have at least $t-1$ keys.

Answer: A

6th variant: (Parallel Algorithm)

1. Four algorithms solve the same problem: polynomial, exponential, logarithmic, linear. Which is fastest?

- A) polynomial
- B) exponential
- C) logarithmic
- D) linear

Answer: C

2. In P-SQUARE-MATRIX-MULTIPLY(A, B), the first 5 lines are:

- 1. $n = A.rows$
- 2. let C be a new $n \times n$ Matrix
- 3. parallel for $i = 1$ to n
- 4. parallel for $j = 1$ to n
- 5. $C[i][j] = 0$

Which continuation is correct?

- A) for $k = 1$ to n ; $C[i][j] = A[i][k] * B[k][j]$; return C
- B) for $k = 1$ to n ; $C[i][j] = C[i][j] + A[i][k] * B[k][j]$; return C
- C) parallel for $k = 1$ to n ; $C[i][j] = A[i][k] * B[k][j]$; return C
- D) parallel for $k = 1$ to n ; $C[i][j] = C[i][j] + A[i][k] * B[k][j]$; return C

Answer: B

3. In LIST-DELETE(L, x), first 3 lines:

- 1. if $x.prev \neq NIL$
- 2. $x.prev.next = x.next$
- 3. else $L.head = x.next$

Which continuation is correct?

- A) if $x.next = NIL$; $x.next.prev = x.prev$
- B) if $x.next \neq NIL$; $x.next.prev = x.prev$
- C) if $x.next = NIL$; $x.next.prev = x.next$
- D) if $x.next \neq NIL$; $x.next.prev = x.next$

Answer: B

7th variant: (P, NP and NP-Complete)

1. A problem is NP-complete if
- A) it is in P and it is in NP
 - B) it can be verified in polynomial time
 - C) it is in NP and can be verified in polynomial time
 - D) it is in NP and it is NP-hard

Answer: D

2. A problem is NP-complete if
- A) it can be solved in polynomial time
 - B) it can be verified in polynomial time
 - C) it can be solved in polynomial time and every other NP problem reduces to it
 - D) it can be verified in polynomial time and every other NP problem reduces to it

Answer: D

3. A problem is NP if
- A) it can be solved in polynomial time
 - B) it can be verified in polynomial time
 - C) it can be solved in polynomial time and every other NP problem reduces to it
 - D) it can be verified in polynomial time and every other NP problem reduces to it

Answer: B

4. A problem is P if
- A) it can be solved in polynomial time
 - B) it can be verified in polynomial time
 - C) it can be solved in polynomial time and every other P problem reduces to it
 - D) it can be verified in polynomial time and every other P problem reduces to it

Answer: A

5. The radix sort(A, d) function is:
- A) for $i = 1$ to d : use a sort to sort A on digit i
 - B) for $i = 1$ to d : use a sort to sort A on digit d
 - C) for $i = 1$ to d : use a stable sort to sort A on digit i
 - D) for $i = 1$ to d : use a stable sort to sort A on digit d

Answer: C

Good luck with the exam

- sadman

1) The average case time searching in a hashtable ?

Ans: $O(1)$

2) The vector properties are?

Ans: static & associative

3) The Binary search tree properly allows us to print out all the keys in a binary search tree in sorted order by a simple recursive algorithm called?

Ans: inorder traversal

4) Nodes have lower and upper bounds on the number of keys they can contain , we express these bounds in terms of a fixed integer t called the minimum degree of a B tree. choose the correct b tree property

Ans: Every node may contain at most $2t-1$ keys

5) Find a false property of a redblack tree

Ans: if a node is black then both children are black

6) The average case time for searching in a linked list is

Ans: $O(n)$

7) We can use the following algorithm and data structure to find the shortest path between two vertices of a directed graph.

Ans: Breadth first search using a queue

8) The problem is NP hard if?

Ans: for every problem in B in NP, there's a polynomial time reduction from B to A

9) A problem is NP complete if?

Ans: it is in NP & it is NP hard

10) A red black tree with n internal nodes has height at most

Ans: $2 \cdot \log(n+1)$

11) Nodes have lower and upper bounds on the number of keys they can contain

We express these bounds in terms of a fixed integer t called the minimum degree of the B tree. choose the correct Btree property

Ans: Every node other than the root must have at least t-1 keys.

12) The queue properties are?

Ans: dynamic and sequential

13) The average case time in a hashtable is ?

Ans: $O(1)$

14) Let us denote the minimum degree of the b tree by t. Find the true sentence

Ans: every internal node of the Btree may have at most $2 \cdot t$ pointers

15) The stack properties are?

Ans: Dynamic and sequential

16) The matrix properties are?

Ans: static and associative

17) We cannot reconstruct the original expression tree from?

Ans: infix

18) The average case time of searching in a red black tree is?

Ans: $O(\log_2 n)$

19) We have four algorithms solving the same problem. First solution runs in polynomial time, second in exp. time , third in log. and last in linear, which is the fastest?

Ans: Logarithmic

Sample Questions

1) The 2 dimensional array is

- A) static and sequential data structure.
- B) dynamic and sequential data structure.
- C) static and associative data structure.
- D) dynamic and associative data structure.

Ans: C

2) The average-case time of searching in a hash-table is

- A) $O(\log n)$
- B) $O(n)$
- C) $O(n*n)$
- D) $O(1)$

Ans: D

3) For each node of a red-black tree, all simple paths from the node to descendant leaves

- A) contain the same number of black nodes.
- B) contain the same number of red nodes.
- C) contain different number of black nodes.
- D) contain different number of red nodes.

Ans: A

4) Nodes have lower and upper bounds on the number of keys they can contain. We express these bounds in terms of a fixed integer $t \geq 2$ called the minimum degree of the B-tree. Choose the correct B-tree property:

- A) Every node other than the root must have at least $2^{*t}-1$ keys.
- B) Every node other than the root may contain at most $2^{*t}-1$ keys.
- C) Every node may contain at most $2^{*t}-1$ keys.**
- D) Every node must have at least $2^{*t}-1$ keys.

Ans: C

5) We have four algorithms solving the same problem. First solution runs in polynomial time, second solution runs in exponential time, third solution runs in logarithmic time and the last solution runs in linear time. Which is the fastest?

- A) polynomial
- B) exponential
- C) logarithmic**
- D) linear

Ans: C

6) A problem is NP complete if

- A) it is in P and it is in NP.
- B) it can be verified in polynomial time.
- C) it is in NP and it can be verified in polynomial time.

D) it is in NP and it is NP-hard.

Ans: D

The depth-first traversal algorithm uses STACK data structure

The breadth-first traversal algorithm uses QUEUE data structure

The average case time of searching in a red-black tree is $O(\log_2 n)$

B-Tree formula: $h \leq \log_t(n+1)/2$

Red-Black Tree: $2 * \log_2(n+1)$

Let us denote the minimum degree of the B-tree by t . **every internal node may contain at most $2*t$ pointers.**

The worst case time of inserting a new element into an AVL tree is $O(\log_2 n)$

Every internal node of the B-tree contains more pointers than keys.